

Kevin Xu

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Professional Summary

AI and Data Engineer with experience architecting high-throughput PySpark/SQL data pipelines, production LLM agent architectures, and PEFT/LoRA fine-tuning workflows. MS in Statistics from UC Berkeley. Skilled in bridging cloud infrastructure with complex statistical modeling to design pre-warehouse validation layers, latent embedding search, and anomaly detection systems. Proven track record taking end-to-end systems from MVP design to high-availability deployment for federal portfolios and large-scale data applications.

Skills

LLM & Agentic AI: LangChain, RAG Systems, Vector Databases (ChromaDB), Parameter-Efficient Fine-Tuning (PEFT/LoRA), SQL AST Parsing & Validation (sqlglot), Human-in-the-Loop Evaluation, Prompt Engineering.

Data & Cloud Engineering: PySpark, Databricks, AWS (Glue, S3, Athena, Step Functions), Data Modeling (dbt), Distributed Systems, ETL/ELT Pipelines, Pre-Warehouse Validation, Schema Enforcement and Deduplication.

Languages & ML Frameworks: Python, SQL (MySQL, T-SQL), PyTorch, Hugging Face, Scikit-Learn, LightGBM, Pandas, NumPy, Statsmodels.

Statistical & ML Methods: Time-Series Forecasting (SARIMA), Feature Attribution (SHAP/LIME), Bayesian Optimization, Anomaly Detection, Spatial Cross-Validation, Regression Analysis.

Developer Tools & Workflow: Docker, FastAPI, RESTful APIs & Webhooks, Git, Bash, Pipeline Profiling & Automated Alerting, Jupyter

Work Experience

Consultant (Data Science)

Deloitte, San Francisco

Aug 2022 - Apr 2025

- Automated Rule Engine & Anomaly Detection:** Designed and owned a rule-based anomaly detection engine across a \$12.2B defense financial portfolio, automating pattern recognition for valuation mismatches, duplicate records, and cross-system breaks. Surfaced ~\$22M in recoverable assets, iteratively feeding audit findings back into system detection logic.
- High-Throughput PySpark Pipelines:** Re-engineered 8+ legacy reconciliation workflows into modular PySpark/Databricks pipelines with explicit schema contracts and execution gates; reduced processing latency from hours to minutes (~80% runtime reduction) under strict EY audit constraints.
- Data Hygiene & S3 Ingestion Framework:** Architected batch ingestion pipelines consolidating multi-source data streams into a unified **AWS S3 Data Lake**. Built pre-warehouse validation layers, schema enforcement, and threshold-based quality gates to prevent corrupted data from propagating downstream.
- End-to-End Data Lineage & Governance:** Engineered a full-graph data lineage framework tagging records with transformation histories and validation outcomes, ensuring pipeline transparency for external audit compliance.
- Observability & Technical Leadership:** Authored system-level technical documentation, built real-time portfolio health dashboards, and mentored 4 junior data scientists on production workflows and pipeline reliability.

Data Science & AI Engineering Projects

Language to Brain Signal Encoding: Fine-tuning BERT with LoRA for fMRI Signal Prediction

- Fine-tuned pretrained BERT models (bert-base-uncased) using **Parameter Efficient Fine-Tuning (PEFT/LoRA)** in PyTorch to adapt contextual language representation to high-dimensional BOLD signal target spaces across 90,000+ voxels.
- Performed hyperparameter tuning across low-rank factorization matrices, learning rates, and target modules, improving validation loss on masked language modeling objectives.
- Evaluated model feature attribution and semantic trigger tokens using game-theoretic interpretability methods (**SHAP, LIME**) across temporal context windows.

Natural Language Text-to-SQL Agent for Analytics

- Engineered an end-to-end Agentic AI Text-to-SQL system using LangChain and LLM reasoning workflows to convert natural language queries into executable optimized SQL for churn and revenue analytics.
- Implemented schema-aware prompt engineering, vector-based semantic retrieval for schema context, and automated error-handling/self-correction loops to ensure production-grade SQL validity.

Polar Cloud Detection via Autoencoders & Ensemble Classifiers

- Implemented transfer learning by pre-training multi-layer neural network **autoencoders** on unlabeled multi-angle satellite imagery to extract 40-dimensional latent embeddings across spatial 3x3 to 9x9 neighbor windows.
- Trained **LightGBM, Random Forest**, and **k-NN** models on multi-angular spatial features, achieving **98.74%** classification accuracy on complex unstructured imagery.
- Conducted spatial cross-validation (quadrant-based) and perturbation/noise injection stress testing to evaluate model stability and out-of-distribution generalization.

Education

University of California, Berkeley

Master of Arts, Statistics | Elizabeth Scott Excellence Scholarship

Sep 2024 - May 2026

University of California, Davis

Bachelor of Science, Statistics | 6x Dean's List

Sep 2020 - Jun 2022